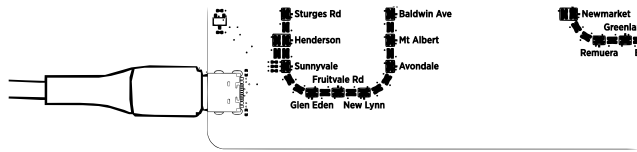


Setup

1. Connect the train map to your computer with the supplied USB Cable.



⚠ For WiFi Setup you must use the left USB-C connector (The right one is power only)

2. Open a browser (e.g. Chrome) and go to: keastudios.co.nz/wt

Firefox and Safari currently don't work because they don't have web-serial support



3. Select Firmware (Auckland V1.1) and Click Connect

4. Allow Serial Port Access

Click on your device and then click Connect in the pop-up to grant device access. If multiple devices appear, look for: **USB JTAG/serial debug unit**

5. Connect to your WiFi Network

Click **Change WiFi** in the LED-Rails Firmware pop-up window and select your WiFi from the drop down and then enter the password for your WiFi.

For the latest Quick Start Guide: keastudios.co.nz/akl-ltm-quick-start/

Care

Avoid exposure to extreme temperatures and abrasive materials like sand.

Never expose the train map to water. If it has been in contact with water do not try to dry it in rice. Keep it powered off and place it in a warm dry place (less than 80°C) for at least a few days before powering it on again.

You can clean the train map using $\geq 95\%$ isopropyl alcohol and a clean lint free cloth (only while the device is unpowered).

If you want to take this map on the go, you can connect it to your phone's WiFi hotspot and use a USB battery bank to power it.

Specifications

Requires $5V \geq 1A$ via USB Type C

Works with both USB C-C and A-C Cables

Supports WiFi 4 (802.11b/g/n, 2.4Ghz)

Data ≈ 0.1 Gigabytes per month

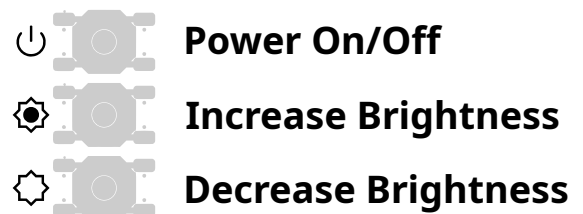
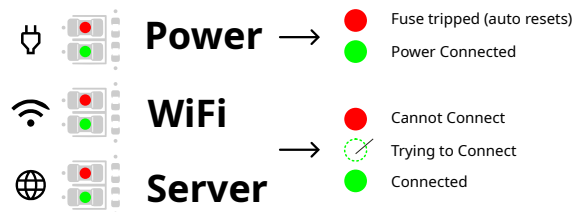
Power $\approx 0.2-0.5kWh$ per month $\approx \$0.07-0.18$

RoHS (Lead Free)

Train Lines



Status Lights and Buttons



Modes

Long press the power button to change modes

1. Realtime (Default)

Fetches the latest train positions from the server every 10 seconds over Wi-Fi

2. Timetable

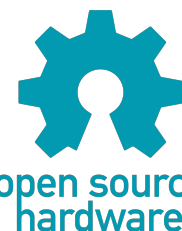
Displays the October 2025 weekday timetable. If it has WiFi, it uses it to display the timetable at the current time of day.

3. Fast Forward

At 1000x normal speed the map displays the October 2025 weekday timetable.



open source
initiative



open source
hardware



RISC-V



AKL-LTM-111-%VAR_num%
You can find the schematics
and design files here:
github.com/CDFER/Auckland-LED-Train-Map

Designed by Kea Studios in Auckland, New Zealand